

# Synopsis

On

**MedGuard AI: *Intelligent Multi-Agent System for Medicine Authentication and Safety Verification***



**Group ID: LYAIA106**

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## 1. Title of the Project

MedGuard AI: Intelligent Multi-Agent System for Medicine Authentication and Safety Verification

## 2. Abstract (150–200 words)

The increasing circulation of counterfeit, expired, and illegal medicines poses a serious threat to public health and patient safety. Traditional verification methods are manual, slow, and require expert knowledge, making them unsuitable for large-scale real-time use. To address this issue, this project proposes **MedGuard AI**, an intelligent multi-agent system designed to automate medicine authentication using multi-modal inputs such as text, images, and barcodes.

The system integrates OCR, barcode decoding, semantic search using FAISS, and AI-based reasoning to verify medicines based on authenticity, expiry status, and legality in the Indian pharmaceutical context. MedGuard AI employs a modular microservice architecture where specialized agents handle data extraction, validation, and decision-making. It also generates user-friendly AI summaries to improve accessibility.

Additionally, the system includes a symptom-based module that provides safe home remedies using a local LLM while ensuring strict medical disclaimers. Designed for deployment across web and mobile platforms, MedGuard AI enhances healthcare safety, improves awareness, and provides a scalable solution for real-time medicine verification.

## 3. Introduction / Background

Medicines are critical to effective healthcare, but the presence of counterfeit, expired, and unregistered drugs in the pharmaceutical supply chain has become a major global concern. These unsafe medicines lead to treatment failures, adverse health effects, and loss of trust in healthcare systems.

Current verification methods are largely manual and require domain expertise, making them inefficient and inaccessible to the general public. With the rise of online pharmacies and complex distribution networks, the risk of fake or illegal medicines reaching consumers has increased significantly.

According to your project work (Phase-I), the system already supports **multi-modal inputs (text, image, barcode)** and performs **expiry and authenticity checks with a dataset of 500+ medicines**.

MedGuard AI aims to solve this problem by providing an automated, AI-powered verification platform that combines OCR, barcode scanning, semantic retrieval, and AI reasoning into a unified system. It is designed to be scalable, user-friendly, and deployable across web and mobile platforms.

## 4. Problem Definition

Counterfeit, expired, and illegal medicines are increasingly entering the pharmaceutical supply chain, posing serious risks to patient health. Existing verification systems are:

- Manual and slow
- Not scalable
- Require expert knowledge

- Do not support image or barcode-based verification

There is a critical need for an **automated, intelligent, and accessible system** that can verify medicines in real time using multiple input formats.

## 5. Objectives

1. To develop an AI-based system for real-time medicine verification.
2. To extract medicine details (name, batch, expiry, etc.) using OCR and barcode scanning.
3. To validate medicines based on:
  - a. Authenticity
  - b. Expiry
  - c. Legal status (India-focused)
4. To implement fast semantic search using FAISS-based RAG.
5. To generate clear, user-friendly AI summaries of medicines.
6. To provide a symptom-based safe home remedy module using a local LLM.

## 6. Scope of the Project

MedGuard AI focuses on **software-based medicine verification** and includes:

- Multi-modal input support (text, image, barcode)
- OCR-based data extraction
- Barcode decoding
- RAG + FAISS semantic search
- Authenticity, expiry, and legality validation
- AI-generated summaries
- Web + Mobile platform support

It is applicable to:

- Consumers
- Pharmacies
- Medical stores
- Healthcare professionals

### Limitations:

- No chemical or lab testing
- Dependent on dataset accuracy
- Advisory system (not a replacement for medical professionals)

## 7. Proposed System

MedGuard AI is a **microservice-based, multi-agent system** that processes medicine inputs through multiple stages:

### Input Layer

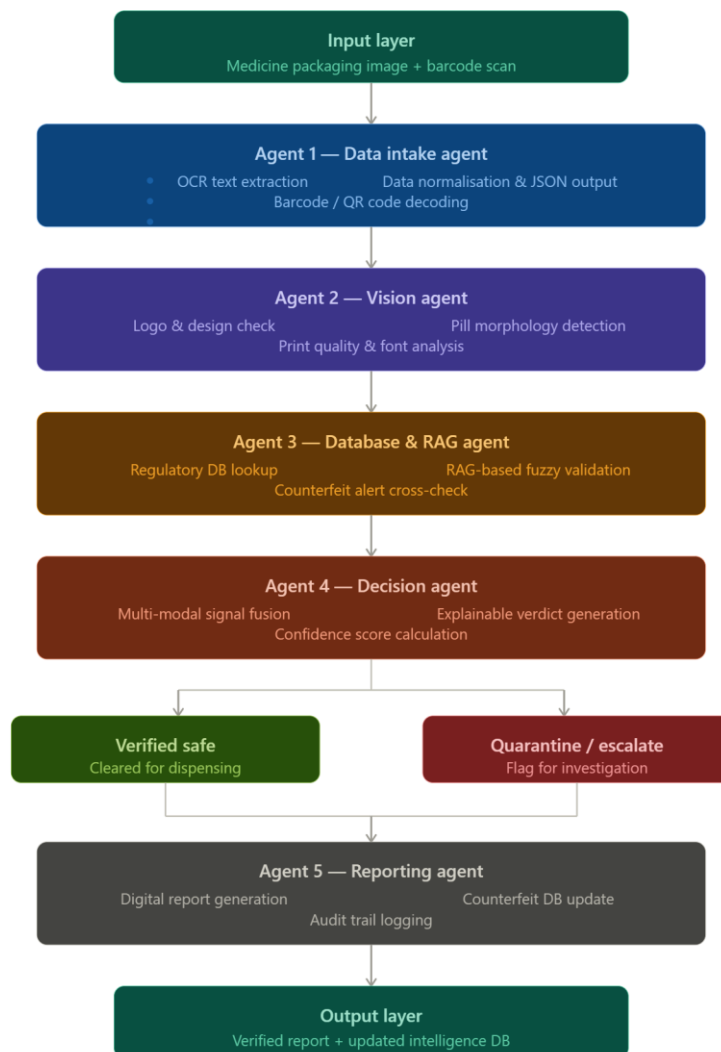
- Text input
- Image upload
- Barcode scan

### AI Processing Layer

1. **OCR Agent** – Extracts text from images (Google Vision API)
2. **Barcode Agent** – Decodes barcode (pyzbar)
3. **Extraction Agent** – Extracts MFG, EXP, MRP using regex
4. **Search Agent** – Performs semantic search using FAISS
5. **Verification Logic** – Determines authenticity, expiry, legality
6. **Summary Agent (LLM)** – Generates readable explanation

### Output Layer

- Authentic / Fake / Expired / Illegal
- AI-generated summary
- Stored verification history



## 8. System Requirements

### a. Hardware Requirements Processor: Intel i5 or higher

RAM: Minimum 8GB (16GB recommended), Storage: 256GB+

## **b. Software Requirements Python**

- Backend: Django, Django REST Framework
- Database: PostgreSQL
- AI & ML: FAISS (Vector Search) , Sentence Transformers , Ollama (Local LLM)
- OCR: Google Vision API
- Barcode: pyzbar
- Frontend: React / Flutter
- Deployment: Docker

## **9. Methodology / Implementation Plan (Project Tracker Sheet)**

The MedGuard AI system follows a phased Agile development approach, ensuring structured progress from planning to deployment. The complete implementation is divided into 10 well-defined phases based on the project development plan. The implementation is divided into multiple structured phases:

- **Phase 1: Requirement Analysis & Planning**  
Defined system scope, requirements, and finalized technology stack. Planned architecture, dataset, constraints, and risk analysis.
- **Phase 2: System Architecture & Database Design**  
Designed microservice architecture and REST communication flow. Created normalized PostgreSQL database schema and system components.
- **Phase 3: Dataset Preparation & RAG Setup**  
Prepared and cleaned medicine dataset with legality and barcode mapping. Generated embeddings and built FAISS index for semantic search.
- **Phase 4: Backend Development**  
Developed API Gateway and core microservices for verification. Integrated OCR, barcode scanning, RAG search, and LLM summary generation.
- **Phase 5: Frontend Development**  
Built React web app and Flutter mobile app for user interaction. Enabled medicine verification, result display, and history tracking.
- **Phase 6: Dockerization & Deployment**  
Containerized all services using Docker for scalability. Deployed system on cloud with persistent storage and secure APIs.
- **Phase 7: Testing & Validation**  
Performed functional, performance, security, and mobile testing. Validated accuracy, response time, and system reliability.
- **Phase 8: Optimization & Hardening**  
Optimized FAISS search, database performance, and system scaling. Improved stability using logging, monitoring, and load handling.

## **10. Expected Outcome**

- A fully functional AI-based system for real-time medicine verification using text, image, and barcode inputs.

- Accurate detection of fake, expired, and illegal medicines with clear classification.
- Fast response system with FAISS-based search (<5 seconds target).
- Generation of user-friendly AI summaries explaining medicine details and safety.
- Secure storage of verification history with scalable web and mobile access.
- Improved patient safety, awareness, and trust in medicine authentication systems.

## **11. Applications / Use Cases**

- Pharmacies & Medical Stores: Verify authenticity, expiry, and legality of medicines before selling to customers.
- Hospitals & Healthcare Providers: Ensure safe medicines are administered to patients.
- Consumers / General Public: Self-check medicines using mobile app for safety and awareness.
- Regulatory Authorities: Assist in monitoring counterfeit and illegal drug circulation.
- Pharmaceutical Supply Chain: Validate medicines during distribution and storage.
- Healthcare Awareness Platforms: Educate users with AI-generated summaries and safe remedy suggestions.

## **12. Future Enhancements/ Future Scope**

- Government API Integration: Real-time validation using official drug databases like CDSCO.
- Pharmacy Dashboard: Bulk verification and compliance monitoring for medical stores.
- Multi-language Support: Support for regional languages like Hindi and Marathi.
- On-device AI Processing: Enable offline OCR and verification for better privacy and speed.

## **13. References**

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